

7. Mobile Application Packaging

Objectives

Checklist to plan your app launch

Prepare for release

App Bundles

Android Go

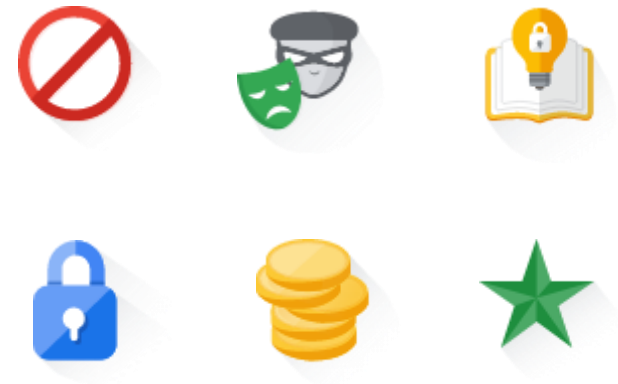
Google Play Instant

Support Chrome OS

Monetize your app

Check List

Developer Program Policies



Developer Program Policies



Restricted
Content



Intellectual
Property



Privacy and
Security



Monetization
and Ads

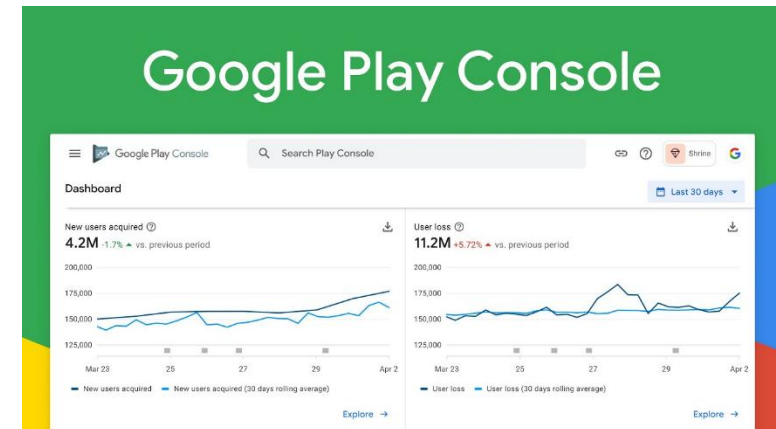


Store Listing
and Promotion

Check List

Developer Program Policies

Developer Account



Developer Account

- It is a **publishing account** issued by **Platform Provider** to **developer**
- It enables developer to **post, display, offer for sale, and distribute apps** through the Platform.

Developer Account



USD 25
(one time)



USD 99
(per year)



FOC

KaiOS

FOC



FOC

Developer Account

Google Play Console

All apps

Inbox 16

Policy status

Users and permissions

Order management

Download reports

Account details

Developer page

Associated developer accounts

Activity log

Setup

Search Play Console

Create app

All apps

View all of the apps and games that you have access to in your developer account

Pinned apps ?

Pin apps here to access them quickly and view key metrics

4 apps

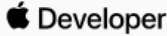
Filter by

All

Search by app or package name

App	Installed audience	App status	Update status	Last updated		
 GBA Market Tracker com.mt.gba.gbamarkettracker	33	Production		6 Oct 2019		
 Multilingual Storybook mmsr.mystorybook	1	Removed by you	 In review	9 Sep 2021		
 TAR UC Bus my.edu.tarc.testing2	0	Removed by you		12 Oct 2017		
 TAR UC Bus Module (Alpha) com.wy.wenyang.tarucbustracking...	0	Removed by you		3 Nov 2016		

Developer Account

Developer

Account

Kwee Teck See ▾

☰ Welcome

Additional Resources

📖 Documentation

⬇ Downloads

💬 Forums

💡 Feedback Assistant

❓ Account Help

📞 Contact Us

Getting Started

Download Xcode, learn how to build an app, and install it directly on your Apple device.



Download Tools

Get started with Xcode, Apple's integrated development environment for creating apps.



Build Your First App

Use Xcode to write a simple "Hello World" app to get familiar with tools, SDKs, and the Swift language.

Join the Apple Developer Program

Membership in the Apple Developer Program includes everything you need to develop, distribute, and manage your apps on the App Store. You'll also gain access to beta software, advanced app capabilities, beta testing tools, and app analytics.

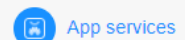
Class Code: 60 15 54

9

Developer Account



Ecosystem services

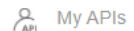


App services



Content services

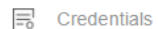
HMS API services



My APIs



API Library

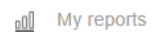


Credentials



Privacy contact info

Developer center



My reports



Customer service



My messages



My accounts



Settings

Distribution and promotion

AppGallery Connect

Your one-stop open platform covers the entire app lifecycle: innovation, development, distribution, operation, and analysis.

Monetize Service



Publisher Service

Development



Push Kit



HUAWEI IAP



HUAWEI ID



HUAWEI HiAI



Health Kit



Wear Engine



HarmonyOS Development Agreement

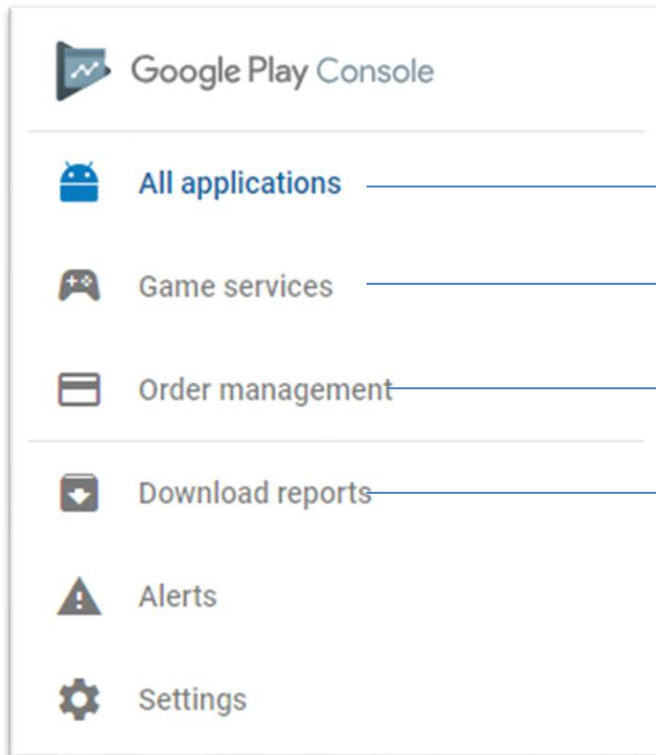


3D Object Recognition

New

Developer Account

Google Play Console - Features



View your apps, active installs, rating, last update and status

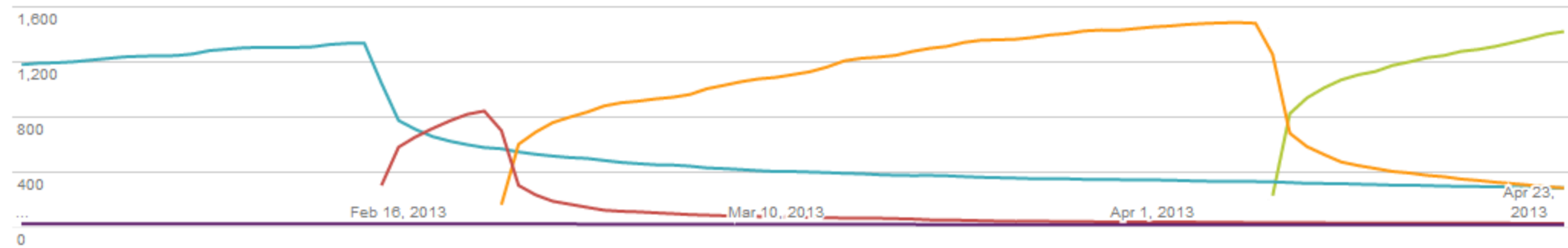
Add social gaming features to your games on Android, iOS and the web

Earn money with paid apps or in-app products

Know your app's performance

Developer Account

ACTIVE DEVICE INSTALLS BY APP VERSION



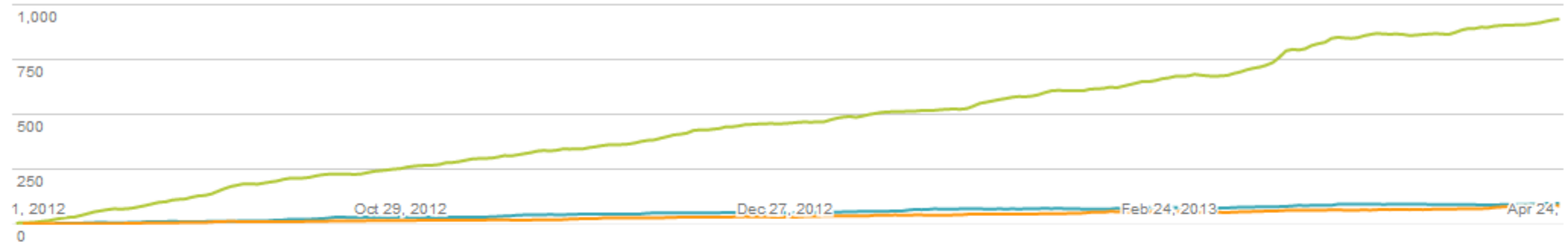
ACTIVE DEVICE INSTALLS ON APR 25, 2013



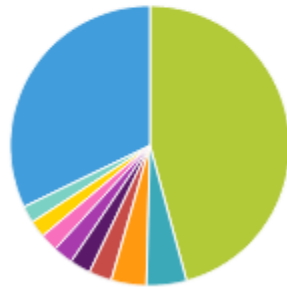
YOUR APP			
☒	36	1,421	69.66%
☒	21	286	14.02%
☒	34	286	14.02%
☒	31	29	1.42%
☒	18	18	0.88%

Developer Account

ACTIVE DEVICE INSTALLS BY COUNTRY



ACTIVE DEVICE INSTALLS ON APR 25, 2013



YOUR APP		
☒ United States	934	45.78%
☒ Canada	94	4.61%
☒ United Kingdom	83	4.07%
☐ South Korea	54	2.65%
☐ Germany	49	2.40%
☐ France	47	2.30%
☐ Australia	42	2.06%
☐ Indonesia	40	1.96%
☐ India	40	1.96%
☐ Others	657	32.21%

ALL APPS IN TOOLS

23.86%
1.33%
2.94%
9.64%
3.88%
2.86%
1.12%
1.10%
2.71%

TOP 10 COUNTRIES FOR TOOLS

United States	23.86%
Japan	11.25%
South Korea	9.64%
Germany	3.88%
Russia	3.06%
Spain	2.97%
United Kingdom	2.94%
France	2.86%
India	2.71%
Brazil	2.52%

Check List

Developer Program Policies

Developer Account

Localization

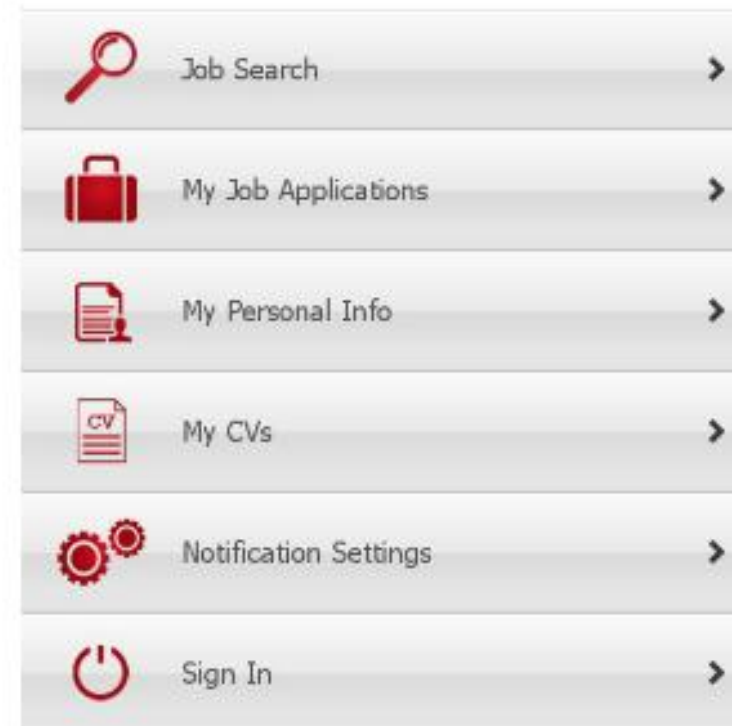
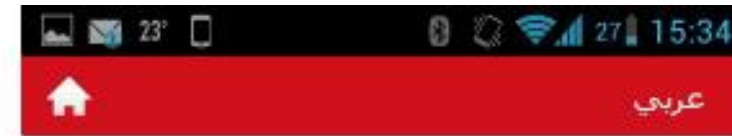


Localization

- The adaptation of an app to meet the needs of a particular **language, culture** or desired population's "look-and-feel."
- A successfully localized app is one that appears to have been developed within the **local culture**.







Check List

Developer Program Policies

Developer Account

Localization

Device Compatibility (Chapter 3)



Device Compatibility

- It ensures that an app runs efficiently **across mobile devices** of **different configurations**.
- Compatibility also should ensure app run well on mobile devices of different **hardware, software, and operating systems**.

Check List

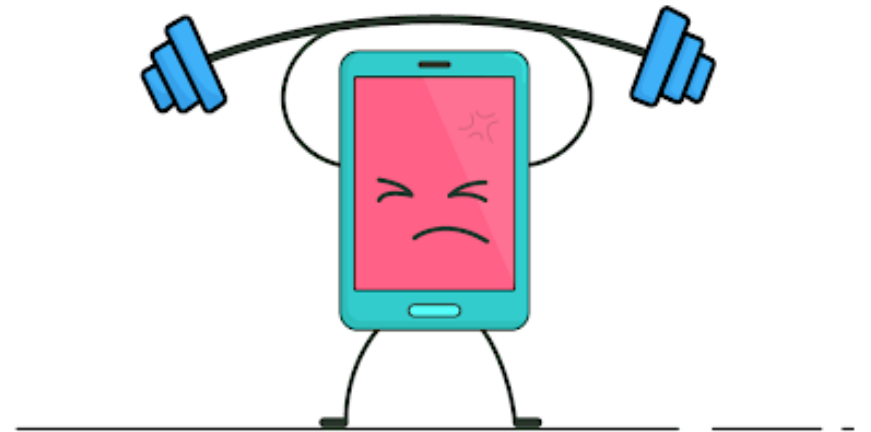
Developer Program Policies

Developer Account

Localization

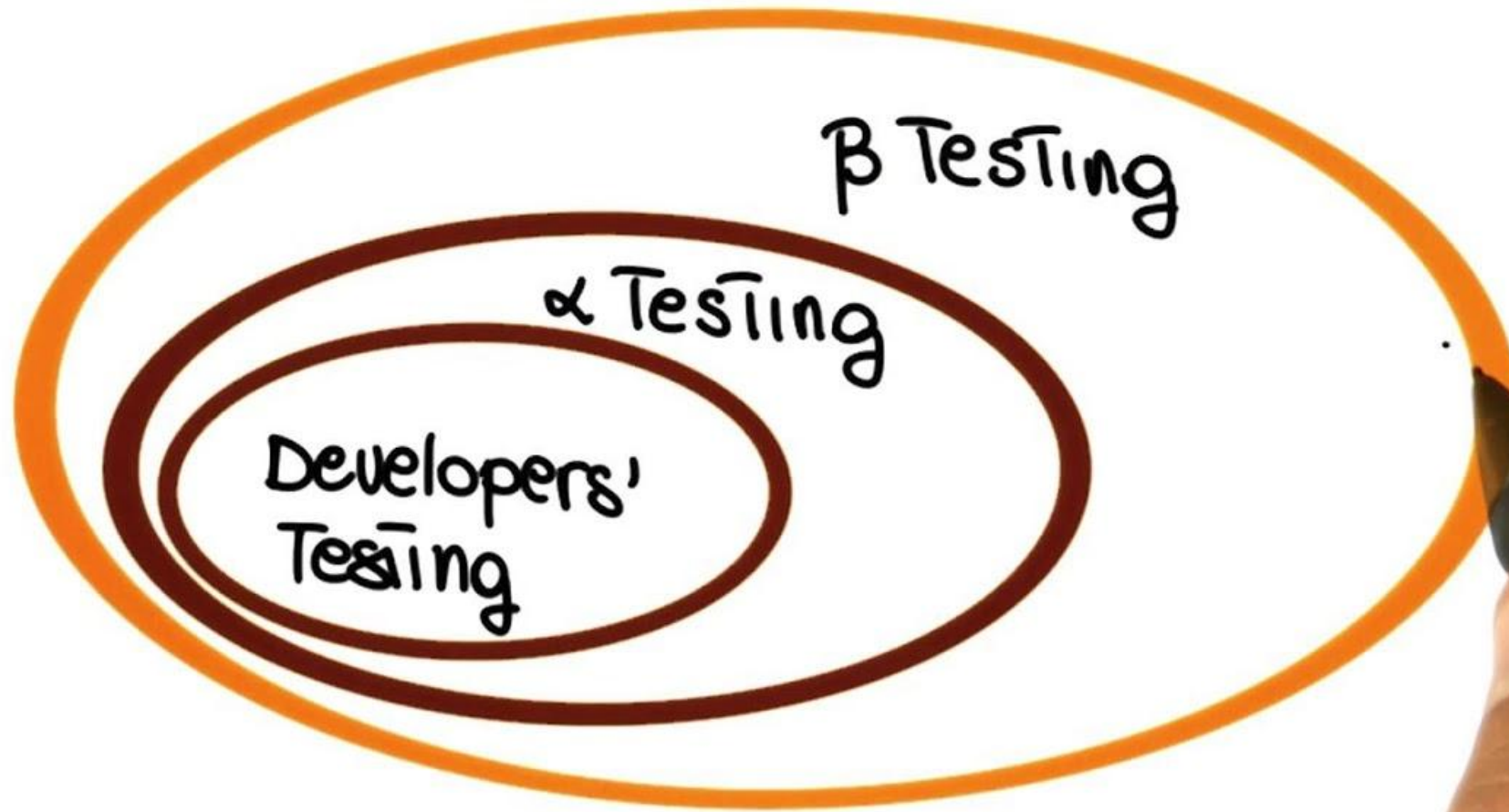
Device Compatibility (Chapter 3)

Quality Test; Alpha & Beta



Quality Test

- A process to ensure apps **meet specified regulations and standards.**
- It's a series of techniques that developers employ to **prevent issues from occurring and ensure they satisfy the customer** with their finished product.



Check List

Developer Program Policies

Developer Account

Localization

Device Compatibility (Chapter 3)

Quality Test; Alpha & Beta

Store Listing



Preparing for Release



To prepare your app so that users can install and run the app on their Android-powered devices

Release-ready .apk file is signed with your own certificate

Preparing for Release

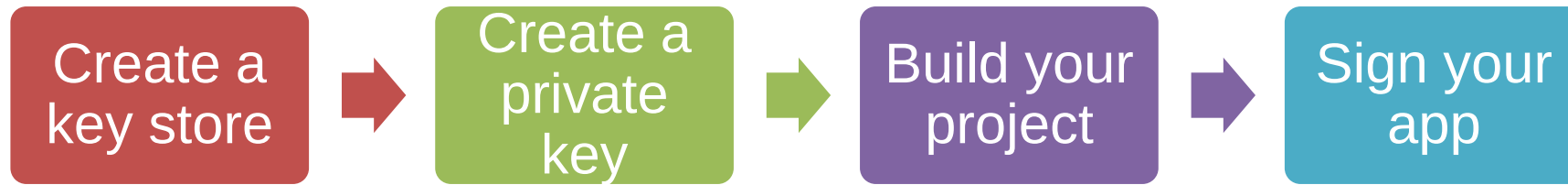
Minimum requirements:

1. Cryptographic keys
2. Application icon
3. End-user License Agreement
4. Promotional and marketing materials

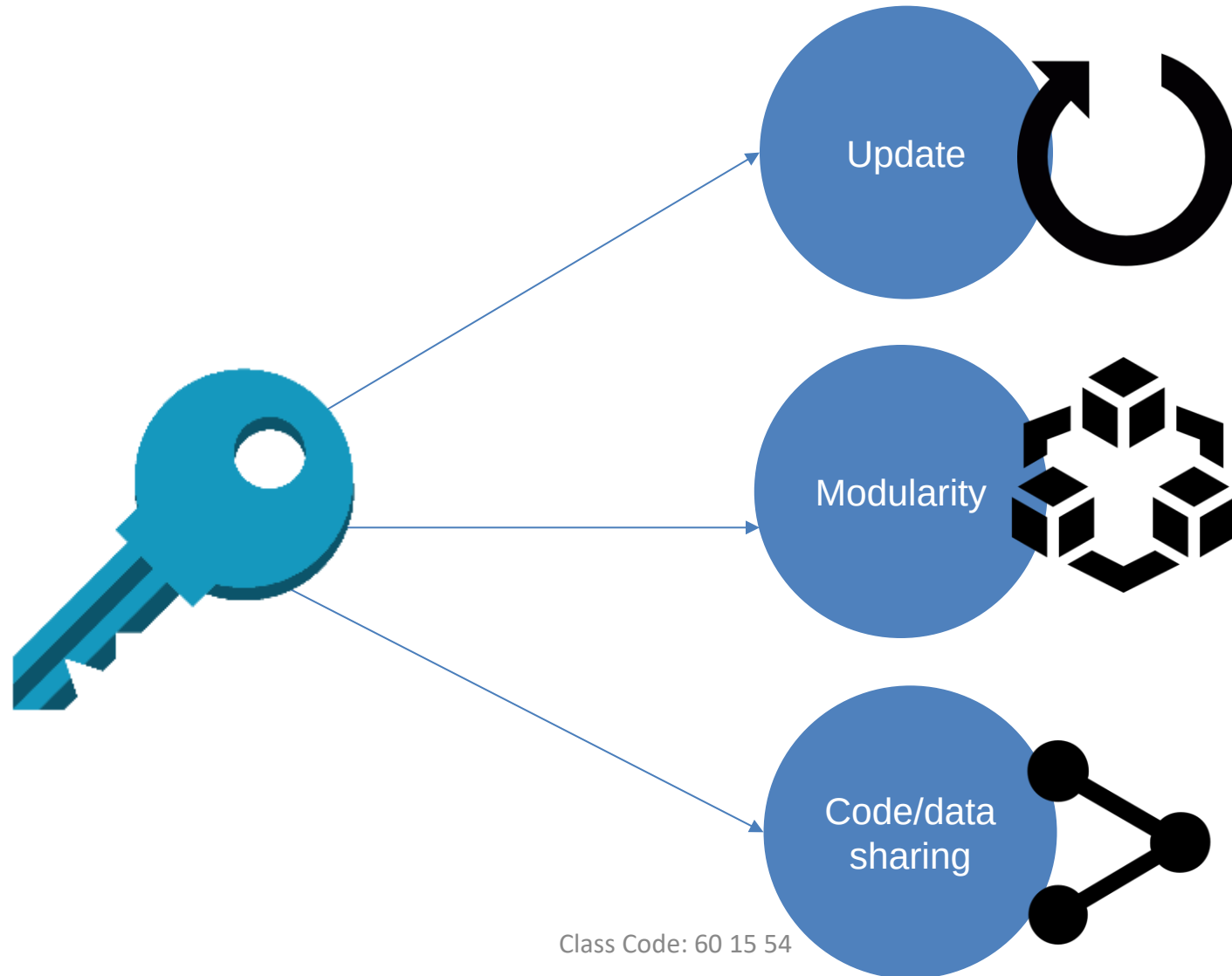


Cryptographic key: digitally signature that is owned by the application's developer

Signing in Release Mode



Signing Consideration



Store Listing



Distribution



App size
<150 MB

+



Expansion
up to 2GB



Platforms

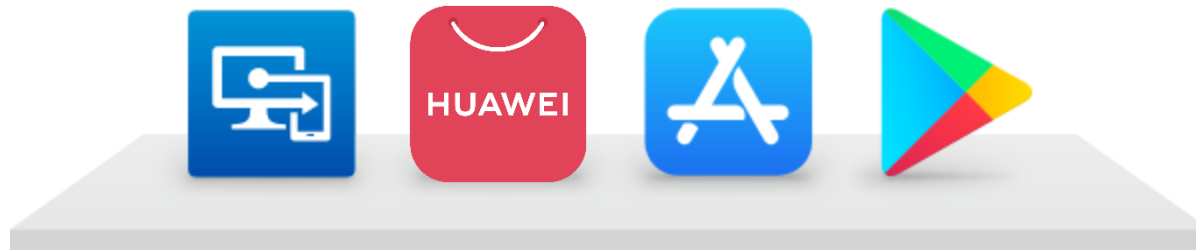


Fee or Free

Question 7.1



Distribution



Distribution Channels



Marketplace



Email



Website or server

Distribution Methods

1. Android Go
2. App Bundles
3. Google Play Instant
4. Support Chrome OS

Android Go

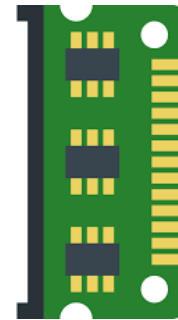
Read the “[Going Global Playbook](#)” (FOC)



Target Oreo
(API 26)



App size less
than 40 MB

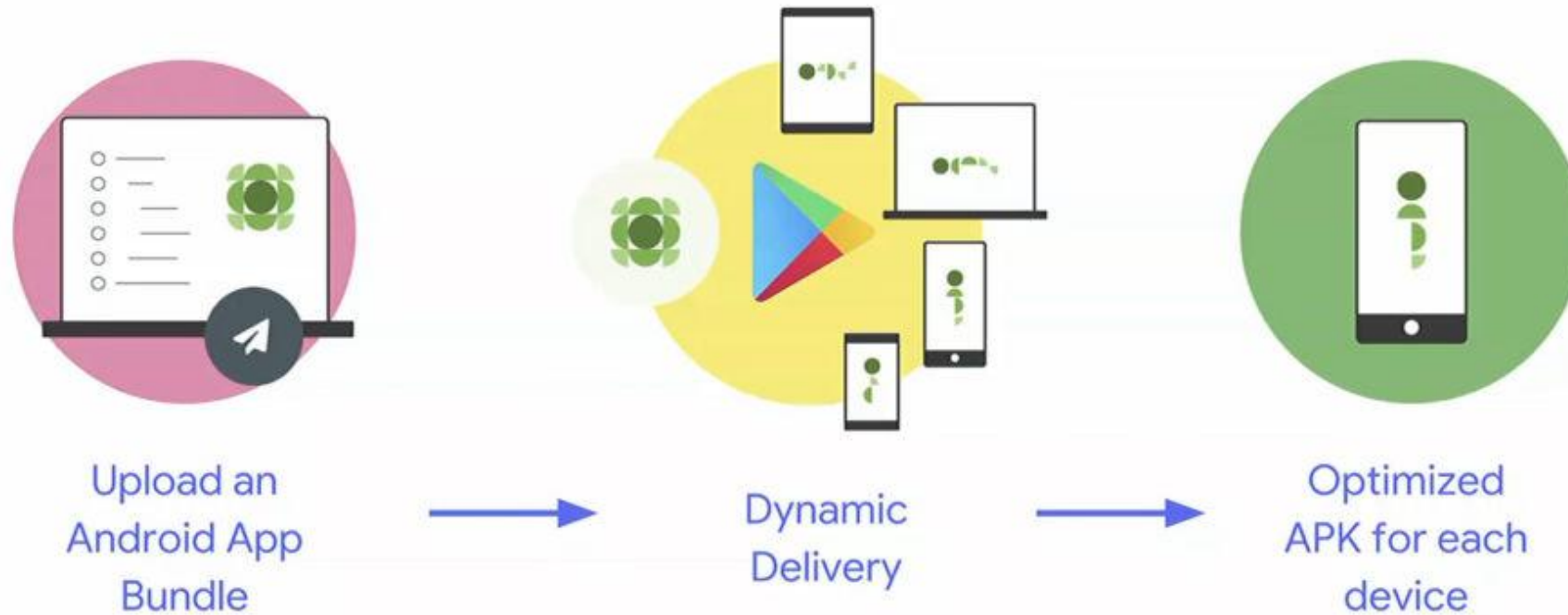


RAM usage
below 50 MB (apps)
150 MB (games)



Start your app
under 5
seconds

App Bundles



App Bundles



App Bundles

Allows developer to upload app's compiled code and resources to Google Play

APK generation and signing of app are deferred

Dynamic Delivery – Google Play uses your app bundle to generate an APK according to user's device configuration

Google Play Instant

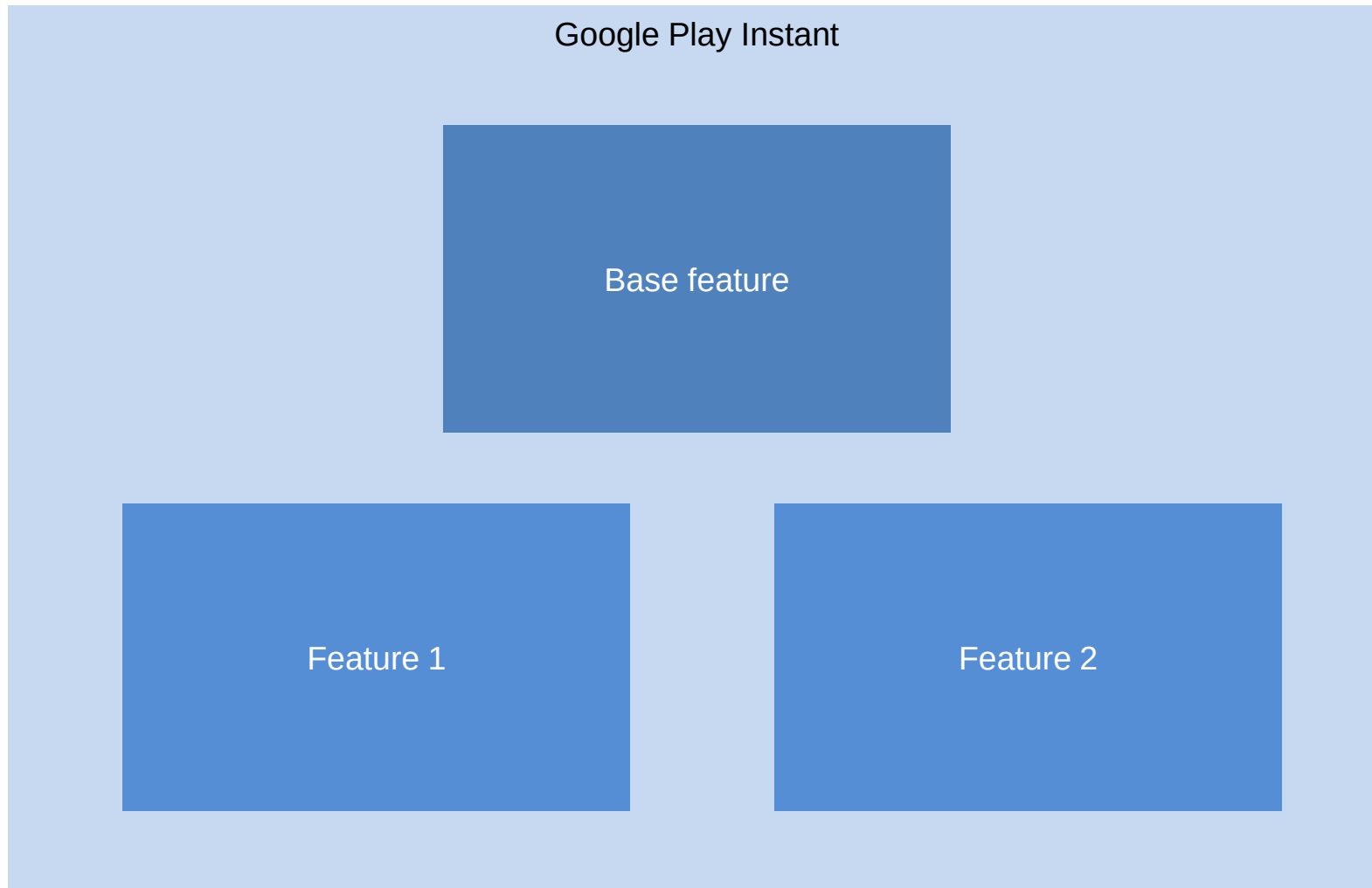
Native Android apps, without the installation

App runs in response to launching a URL

Upgrade to your existing Android app, not a new, separate app

Android 5.0 (API level 21) and above

Supported by App Bundles



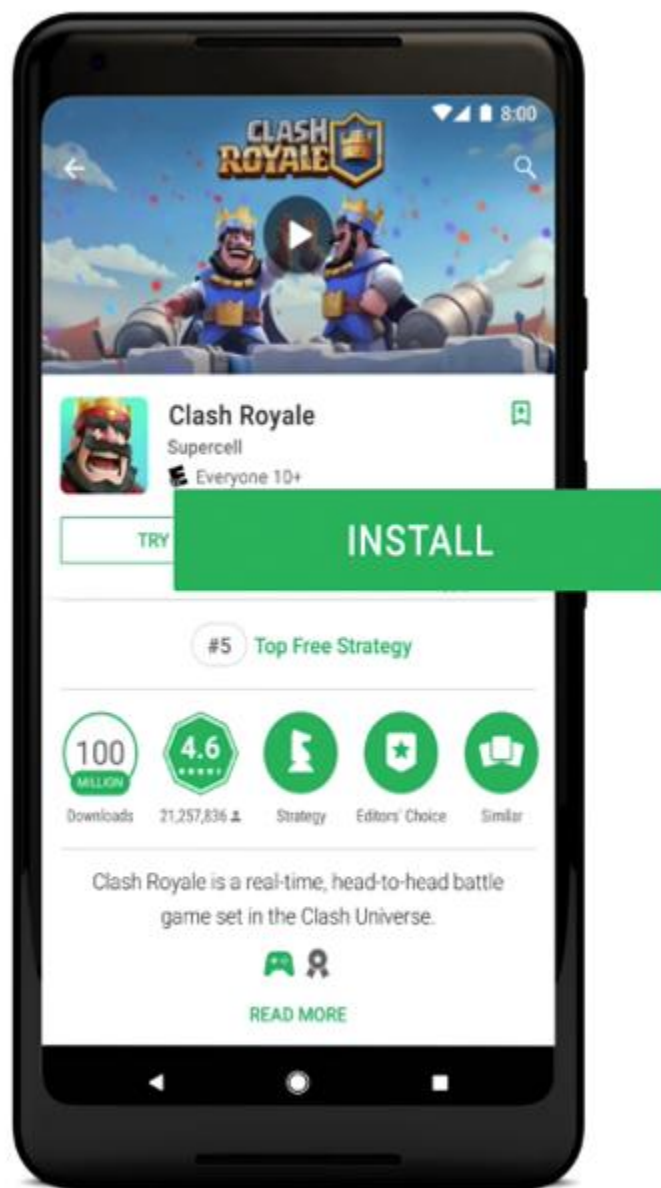
<https://developer.android.com/topic/instant-apps/prepare.html>

Google Play Instant

Two categories:

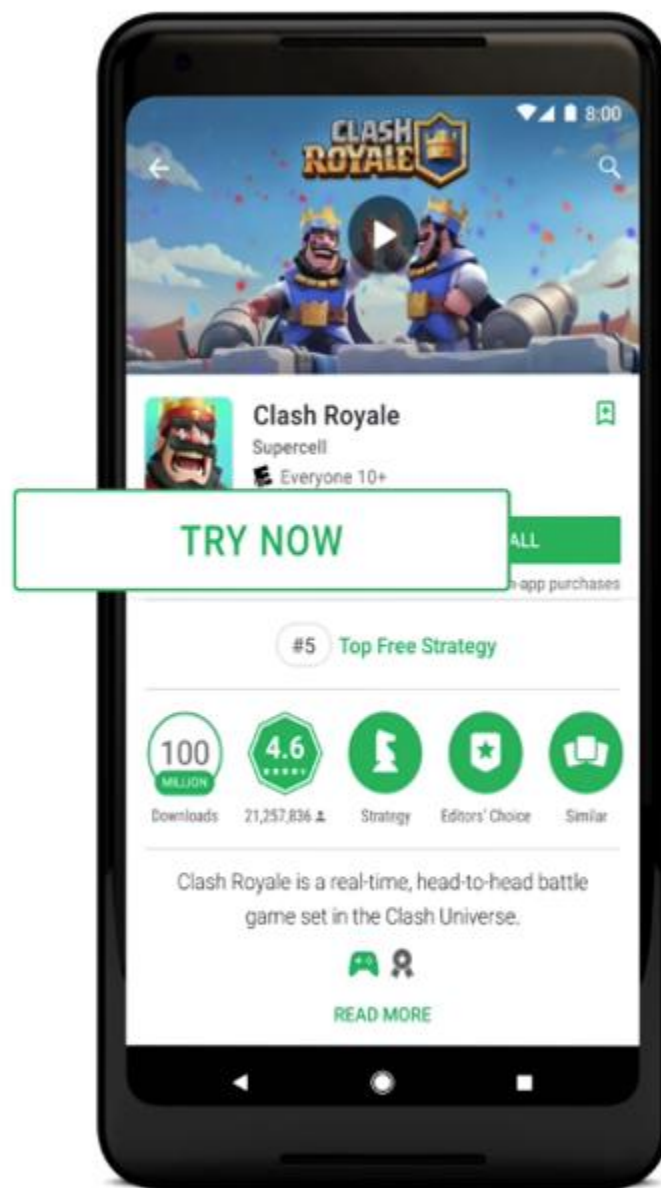
1. Try : experience in the Play Store
2. Instant Play: full game experience in the Play Games app

Limitation : app size is ≤ 15 MB



Download

3



Instant

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Disadvantages of Google Play Instant

- Instant apps can use only a subset of APIs
- App size limitation

Chrome OS



Chrome OS

It supports the Google Play Store and Android apps

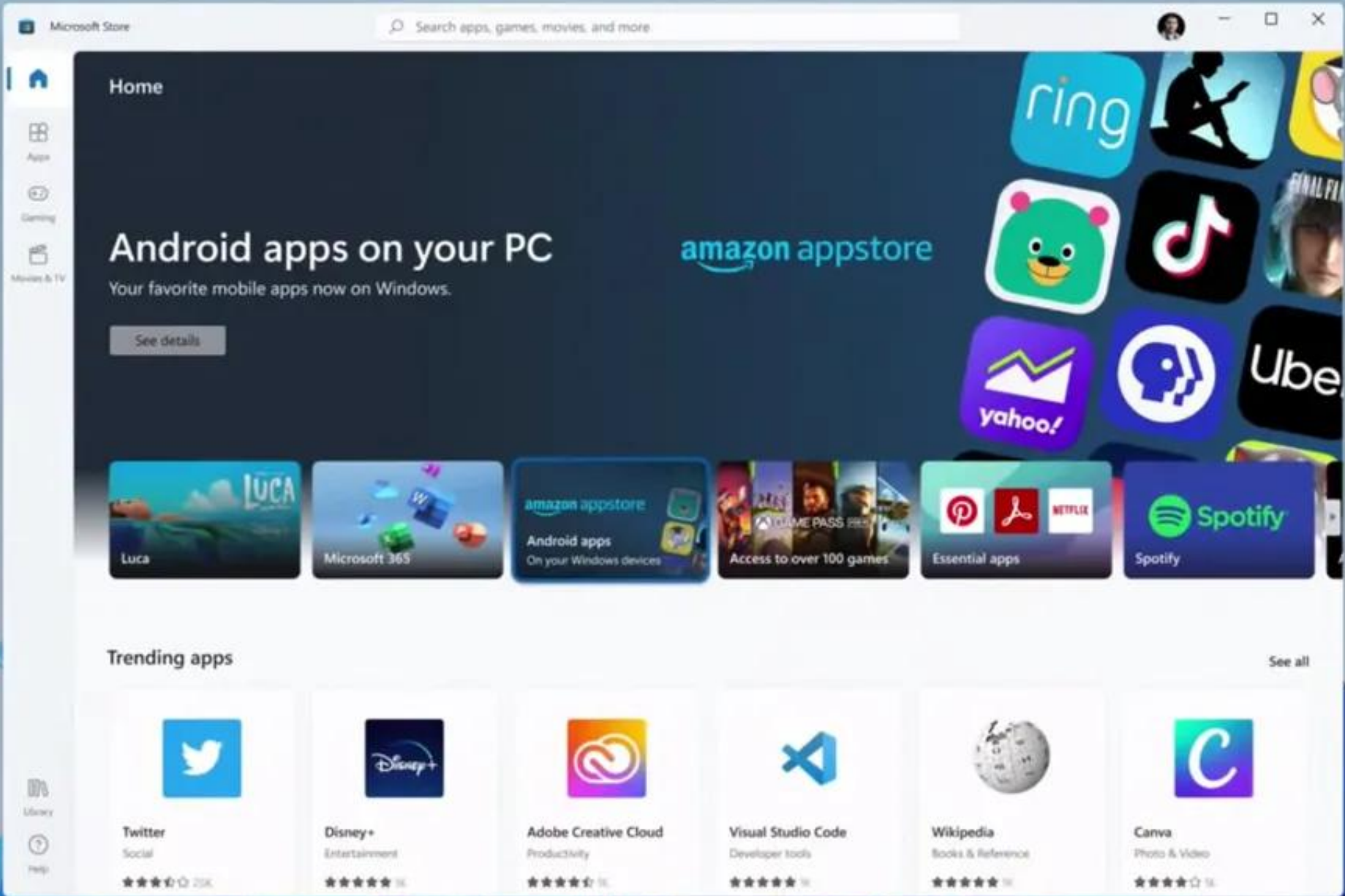
Preparation:

- Manifest file

```
<uses-feature  
  android:name="android.hardware.touchscreen"  
  android:required="false" />
```

Use apps from mobile on PC

Microsoft Store is hosting Android apps and games



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Microsoft's Link to Windows app allows user to access Android apps right on PC



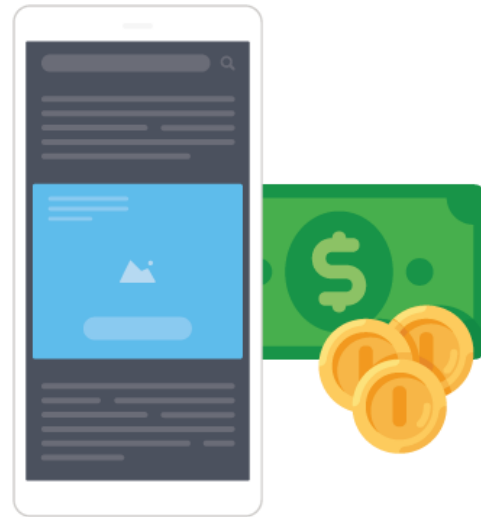
iPhone apps and iPad apps are available on the Mac App Store on Apple silicon Macs without modification



Question 7.2



Monetize Your App



Monetize Your App

Premium Apps

Paid

In-app Billing

Extensive Features

Narrow niche in the market



Monetize Your App

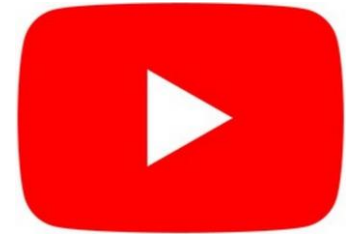
Freemium Apps

Free

Uses In-app Billing

Digital goods

- Durable
- Consumable



Monetize Your App

Subscription

Free trial

In-app billing

Subscription fee



Monetize Your App

Ads

Free

AdMob + Google
Mobile Ads SDK

Shows income-
generating ads



Alternative Monetization Options

E-Commerce

Free

B2C

Technology, logistics
and payment solutions

Sales commissions +
Setup fees



Alternative Monetization Options

Rewarded Products

Free

Share content on social media, perform simple tasks (e.g. scan QR code, answer survey and etc)

Get free stuff, earn rewards



8coin



GigaGigs



Google Opinion Rewards

Alternative Monetization Options

Service

Free

An extension of
physical/online
services



Alternative Monetization Options

Data Collection

Free

Partnership/Affiliate

Provide customer service, promotion, run a giveaway and etc



Question 7.3 Monetize Your App



Question 7.3 Monetize Your App

What is the monetizing model suitable for each of the following app?



a



b



c



d

Review Questions

1. What are the benefits of Instant App?
2. What is the main benefit of the Dynamic Delivery?
3. A junior developer has encountered a problem in distributing a mobile app. The app was developed to cater for both phones and high resolution tablets. As a result, the app contains both normal and high resolution art works that contributes to a large size app, which is currently more than 100 MB. Suggest a suitable solution to solve this problem and justify your answer. (5 marks)

Conclusion

Checklist to plan your app launch

Prepare for release

App Bundles

Android Go

Google Play Instant

Support Chrome OS

Monetize your app